UdbhavaX'26

- **Project Title** : Real-Time Fraud Detection in Digital Payment
  System Domain : FT - 01 (FinTech)
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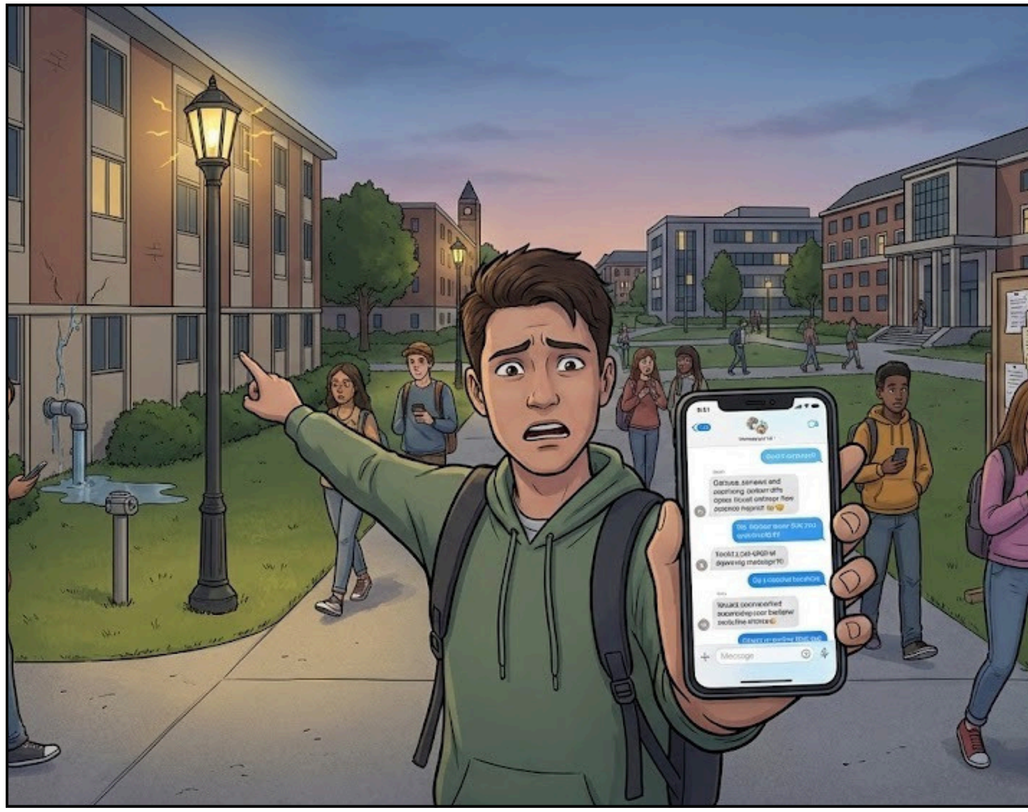
- **Event** : Udbhava X’26, Coding Club SR University

Problem Scenario

The Real-World Situation : Real-time fraud detection is essential in digital payments, where millions of transactions occur every second. Traditional systems often fail to detect fraud quickly, causing financial loss and reduced trust. A smart real-time system can instantly identify and prevent suspicious activities, ensuring secure and reliable transactions.

The Current Failure : In real-time fraud detection is the reliance on outdated and slow systems that cannot detect advanced fraud instantly. As a result, suspicious transactions go unnoticed, leading to financial losses and reduced user trust.



Current Pain Points (Why it Breaks)

- **No Centralized Data Monitoring:** Transaction data is scattered across multiple systems, making it difficult to detect fraud patterns effectively.
- **Duplicate Transaction Alerts:** The same suspicious activity may trigger multiple alerts without proper filtering, wasting time and reducing efficiency.
- **Inability to Prioritize High-Risk Transactions:** Critical fraudulent activities may get lost among low-risk alerts due to lack of proper risk classification.
- **No Historical Data Analysis:** Systems fail to use past transaction data effectively, making it hard to predict and prevent future fraud patterns.

Objectives & Success Criteria

Primary Objective: To develop an intelligent real-time fraud detection system that monitors digital transactions, detects suspicious activities instantly, and ensures secure and reliable payment processing.

Success Criteria:

- Successfully capture and analyze transaction data (User Behavior, Transaction Amount, Location, Time).
- Implement "Smart Logic" using AI/ML to detect and prevent fraudulent transactions in real time.
- Provide real-time alerts and transaction status (Approved / Flagged / Blocked).
- Enable financial institutions to quickly identify and prioritize high-risk transactions for immediate action.

Proposed Solution Strategy

Key Solution Strategies:

1. Data Standardization:

Ensuring structured and consistent transaction data (amount, location, time, user behavior) for accurate fraud analysis.

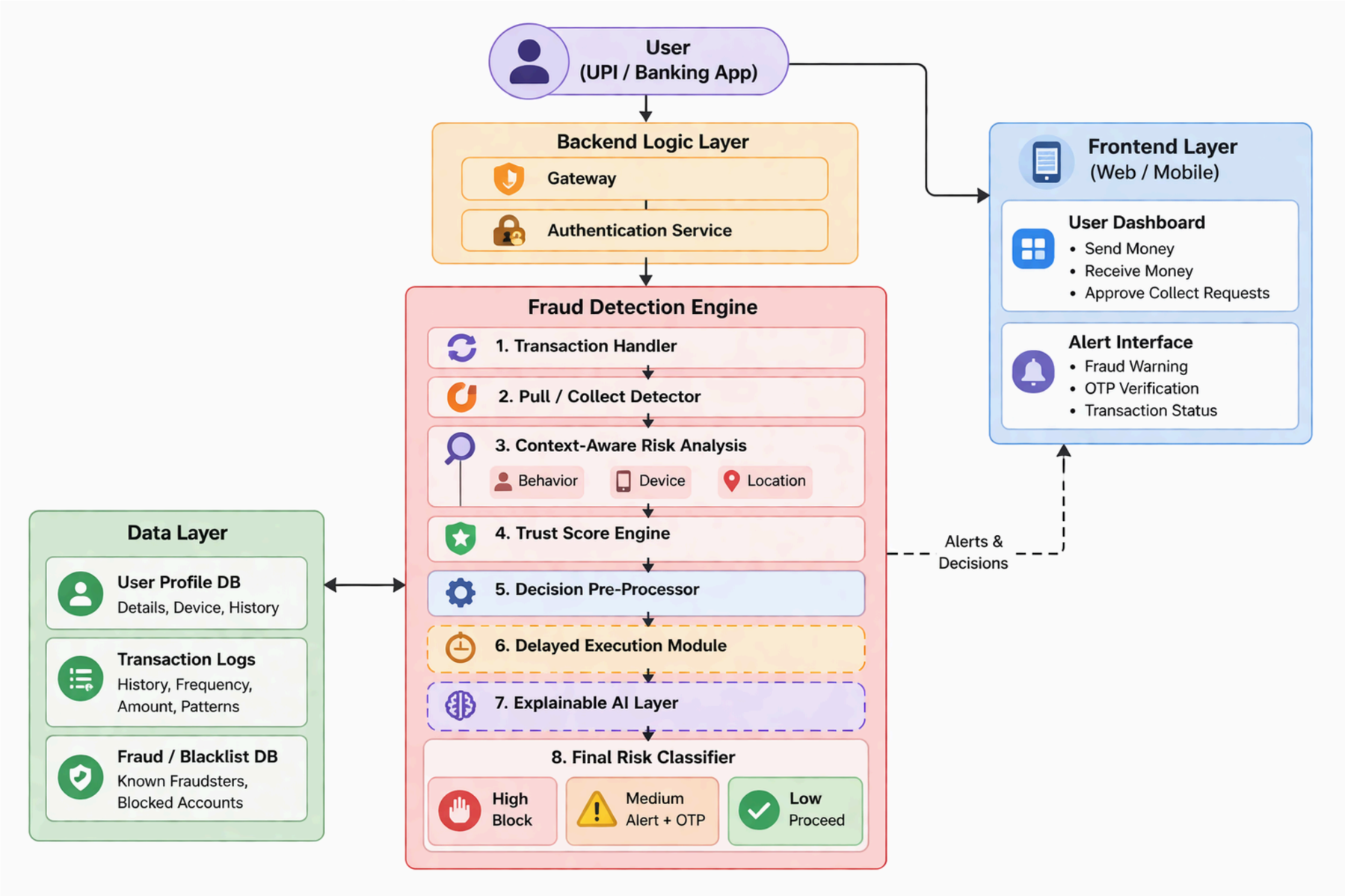
2. Role-Based Access:

Providing different interfaces for users (secure transactions) and administrators (fraud monitoring and management).

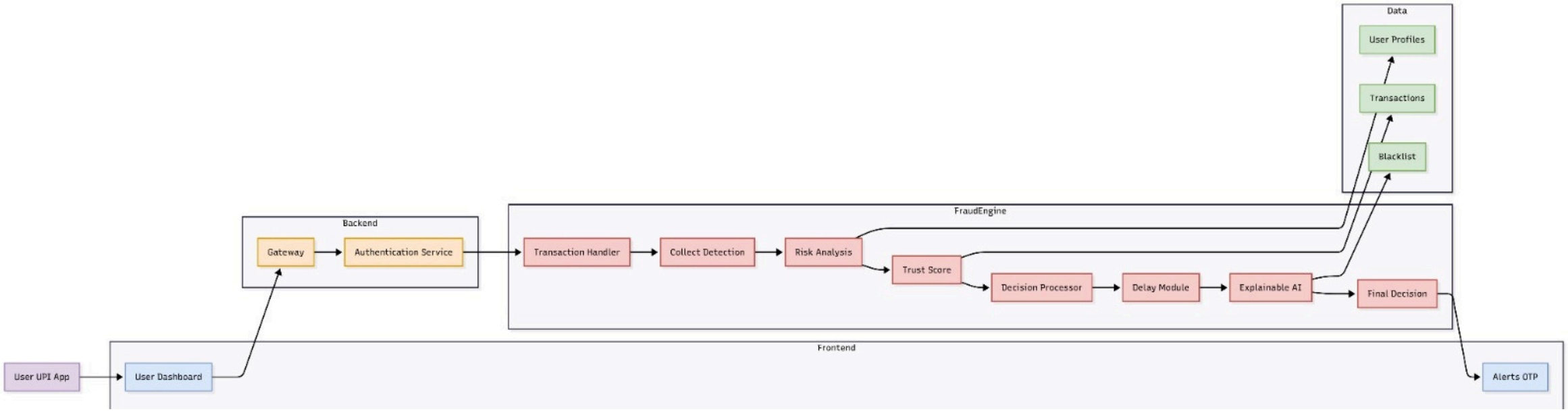
3. Smart Fraud Detection Logic:

Implementing AI/ML-based algorithms to identify, flag, and prevent suspicious transactions in real time.

System Architecture



Data Flow / Workflow



Data Flow / Workflow

1. Submission: Student selects Location/Category and submits.
2. Validation Logic: System checks active records. Decision: If match found Stop & Alert. If unique Proceed.
3. Visualization: Admin Dashboard updates immediately. High-priority items are highlighted visually (e.g., Red).
4. Resolution: Admin updates status to "Resolved." This change automatically reflects in the Student's "My History" view.

Implementation & Demo Reference

Implemented Features (Reference Prototype):

- **Role-based Access (User/Admin):** Secure login system for users and administrators with different access levels.
- **Real-Time Transaction Monitoring:** Continuous tracking of digital transactions to detect suspicious activities instantly.
- **Fraud Detection Dashboard:** Admin dashboard with analytics and statistics for monitoring fraud trends.
  Risk-Based Alert System: Visual alerts (e.g., High-Risk flags) to highlight suspicious or potentially fraudulent transactions.

GitHub : https://github.com/KairamAkshay/FraudX/tree/main

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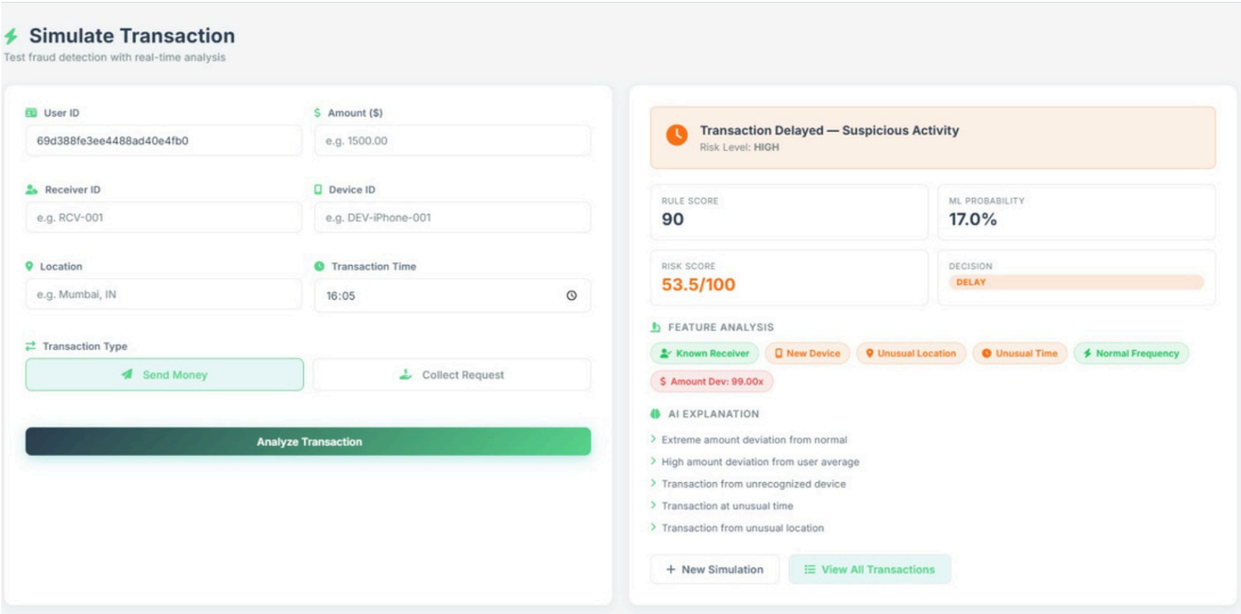
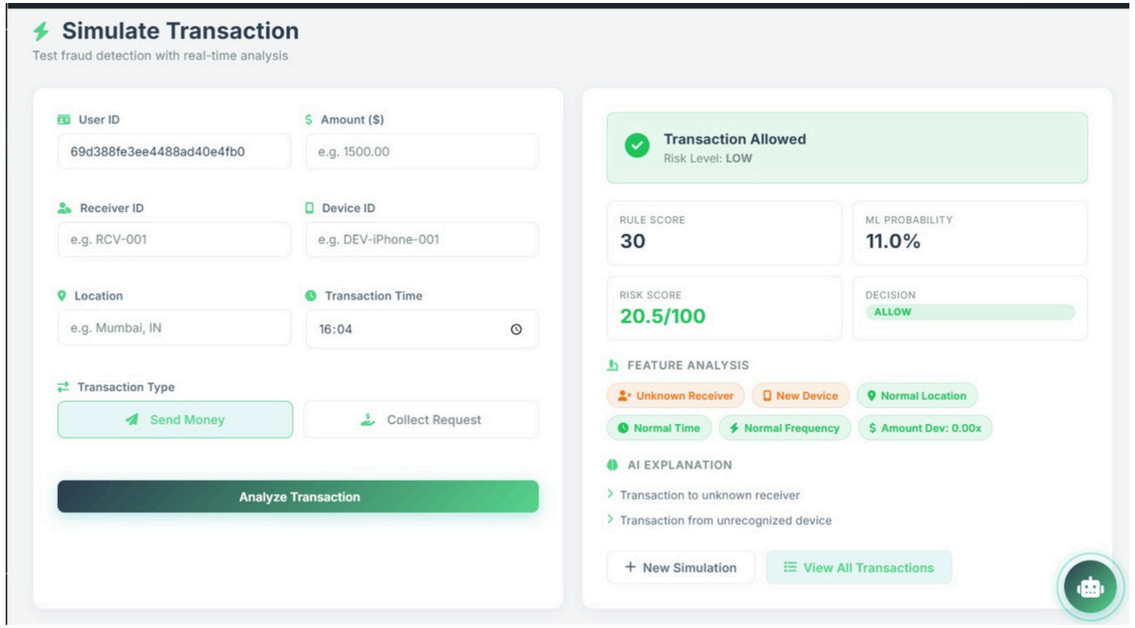




Impact & Benefits

For Users:

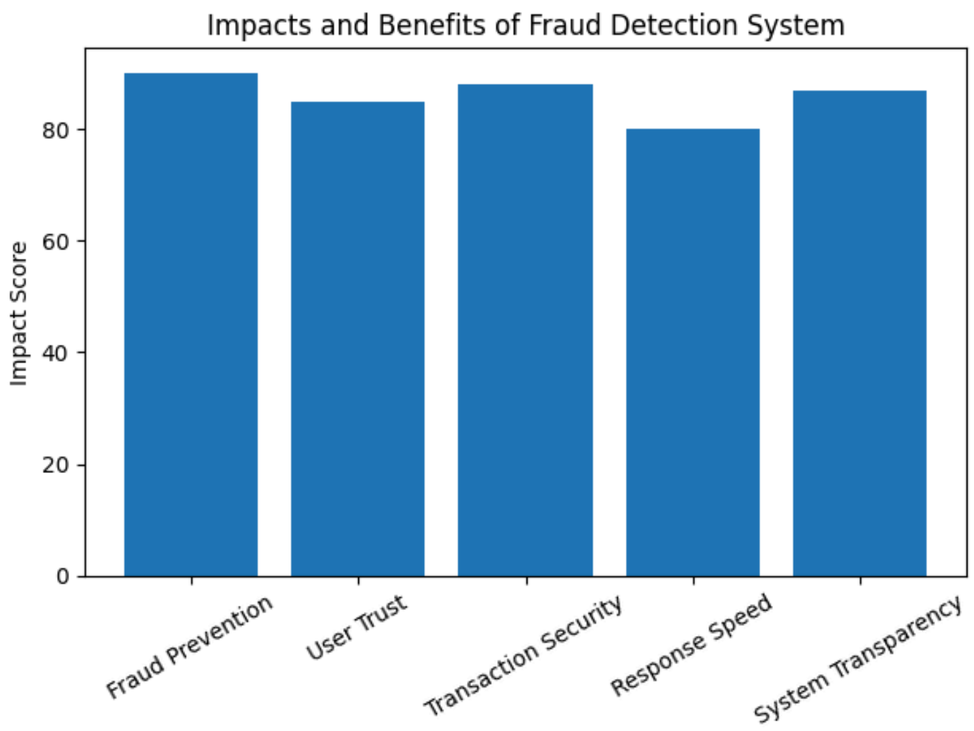
- Enhanced security and trust in digital transactions. Reduces the risk of financial loss and ensures safe payment experiences.

For Financial Institutions:

- Faster detection and prevention of fraudulent activities. Reduces manual effort and improves efficiency in risk management.

For the Organization:

- Access to valuable transaction data for better fraud analysis and decision-making. Strengthens system reliability and builds customer confidence in digital payment platforms.



Limitations

Prototype Data Handling :
For demonstration purposes, the system may use sample or limited datasets instead of real-time large-scale financial data. In production, it requires integration with secure banking databases and cloud infrastructure.

Authentication :
The system uses basic or simulated user authentication and does not fully integrate with advanced banking security systems like multi-factor authentication or biometric verification.

Real-Time Processing Constraints :
While the system detects fraud in near real-time, handling extremely high transaction volumes with zero delay requires more advanced infrastructure and optimization .

Future Scope

Phase 2: AI Chatbot Assistance

Focus: User support and guidance

Provides real-time help, answers queries, and assists users in verifying suspicious transactions

Improves financial awareness and reduces fraud through interactive guidance

Phase 3: Integration with UPI Applications

Focus: Real-world implementation in existing payment systems

Integrates fraud detection features directly into existing UPI apps for real-time monitoring and prevention

Enhances security within the app using risk analysis, alerts, and user verification mechanisms